THE JOURNAL OF PHYSICAL CHEMISTRY B

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Molecular Dynamics Simulations of Micelle Properties and Behaviors of Sodium Lauryl Ether Sulfate Penetrating Ceramide and Phospholipid Bilayers

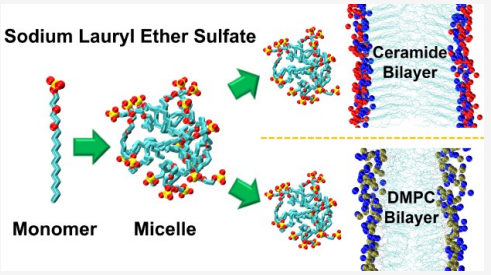
Published as part of The Journal of Physical Chemistry virtual special issue "Peter J. Rossky Festschrift".

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Cite This: *J. Phys. Chem. B* 2020, 124, 5919−5929 Read Online

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ABSTRACT: Molecular dynamics (MD) simulations with the umbrella sampling (US) method were used to investigate the properties of micelles formed by sodium lauryl ether sulfate with two ether groups (SLE2S) and behaviors of corresponding surfactants transferring from micelles to ceramide and DMPC bilayer surfaces. Average micelle radii based on the Einstein−Smoluchowski and Stokes−Einstein relations showed excellent agreement with those measured by dynamic light scattering, while those obtained by evaluating the gyration radius or calculating the distance between the micelle sulfur atoms and center of mass overestimate the radii. As an SLE2S micelle was pulled down to the ceramide bilayer surface in a 400 ns constant-force steered MD (cf-SMD) simulation, the micelle was partially deformed on the bilayer surface, and several SLE2S surfactants easily were partitioned from the micelle into the ceramide bilayer. In contrast, a micelle was not deformed on the DMPC bilayer surface, and SLE2S surfactants were not transferred from the micelle to the DMPC bilayer. Potential of mean force (PMF) calculations revealed that the Gibbs free energy required for an SLE2S surfactant monomer to transfer from a micelle to bulk water can be compensated by decreased Gibbs free energy when an SLE2S monomer transfers into the ceramide bilayer from bulk water. In addition, micelle deformation on the ceramide bilayer surface can reduce the Gibbs free energy barrier required for a surfactant to escape the micelle and help the surfactant partition from the micelle into the ceramide bilayer. An SLE2S surfactant partitioning into the ceramide bilayer is attributed to hydrogen bonding and favorable interactions between the hydrophilic surfactant head and ceramide molecules, which are more dominant than the dehydration penalty during bilayer insertion. Such interactions between surfactant and lipid molecule heads are considerably reduced in DMPC bilayers owing to dielectric screening by water molecules deep inside the head/tail boundary between the DMPC bilayer. This computational work demonstrates the distinct behavior of SLE2S surfactant micelles on ceramide and DMPC bilayer surfaces in terms of variation in Gibbs free energy, which offers insight into designing surfactants used in transdermal drug delivery systems and cosmetics.



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INTRODUCTION

The human skin functions as a barrier by protecting the body from the invasion of various dangerous foreign substances and microbes and by preventing water and vital components from escaping. In particular, the "stratum corneum" (SC), i.e., the outermost epidermal layer, functions as a primary barrier and external environmental interface.[1,2] The SC is composed of terminally differentiated keratinocytes called "corneocytes" and an intercellular matrix of skin lipid molecules, and the SC structure is often described as a "bricks (corneocytes) and mortar (lipids) model".[3] Unlike cell membranes mainly composed of charged or dipolar lipid molecules such as 1,2-dimyristoyl-*sn*-glycero-3-phosphocholine (DMPC), the main intercellular matrix components are non-dipolar ceramide lipids (Figure 1a and b). The SC intercellular lipid matrix directly interacts with exogenous substances and provides the main route for substance permeation through skin.[4] There have been numerous experimental and computational studies on the properties of the intercellular lipid region and its interactions with small molecules.[3,5−15]

Surfactants as a solubilizer can increase the solubility of oils in water through dispersion or emulsification and have long been used in medicine and cosmetics.[16−18] Furthermore, surfactants can enhance drug transport through skin. Consequently, they have served as penetration enhancers for transdermal drug delivery.[19] Undesirable surfactant side effects include the swelling and irritation of contacted skin.[8,17,20]

Received: March 31, 2020
Revised: June 8, 2020
Published: June 17, 2020



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https://dx.doi.org/10.1021/acs.jpcb.0c02856
J. Phys. Chem. B 2020, 124, 5919−5929

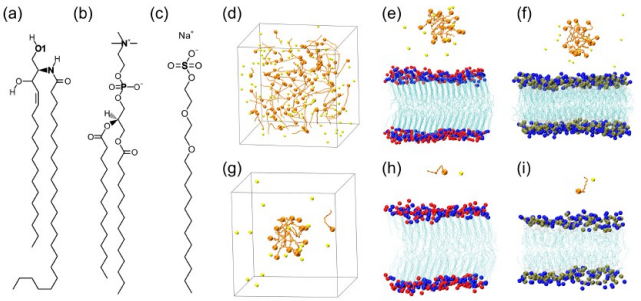


Figure 1. Chemical structures of (a) ceramide (Cer240) and (b) DMPC lipid molecules and (c) SLE2S surfactant and (d–i) initial configurations of six simulation systems. In parts e, f, h, and i, ceramide and DMPC bilayer surfaces are normal to the *z* direction. In parts d–i, sulfur atoms in SLE2S surfactants are highlighted by orange spheres while other atoms are indicated by orange tubes, and sodium cations are shown by yellow spheres. In parts e, f, h, and i, nitrogen (N), oxygen (O1), and phosphorus (P) atoms in lipid-heads are represented by blue, red, and tan spheres, respectively, and lipid tails are cyan. Water molecules are removed for clarity.

Previously, it was believed that only monomeric surfactants could penetrate and damage the skin.[21,22] However, recent studies have revealed that micellar surfactants can cause harm to the skin in a similar way.[8,21,22] Although much effort has been devoted to unraveling surfactant skin-penetration and skin-irritation mechanisms, the precise surfactant mechanism of action remains unclear. Sodium lauryl ether sulfate (SLES, also known as sodium laureth sulfate) is an anionic surfactant showing a hydrophilic head composed of sulfate and oligomeric oxyethylene (OE) units and a hydrophobic hydrocarbon tail (Figure 1c).[17,23,24] Compared to sodium dodecyl sulfate (SDS), which previously had been widely used, SLES is less toxic to the human body and can be mass-produced inexpensively, thereby popularizing it in cosmetic manufacturing.[17,23,25] However, despite its wide application to cosmetics, SLES also irritates skin through a still-unidentified mechanism of action.[17,23,25,26]

Owing to rapid advances in computer technology, molecular dynamics (MD) simulations have become one of the most powerful computational tools for investigating and predicting structural and dynamic properties of complex systems such as biological membranes interacting with small molecules at an atomistic level.[27−29] Using ceramide bilayers to model the SC intercellular lipid matrix, numerous MD simulation studies have explored SC structural and thermodynamic properties. Furthermore, reliable force fields recently developed for skin lipids enable us to examine interactions and behaviors of small molecules penetrating skin.[3,7,14,30−33] For instance, Guo et al. developed a set of force fields for ceramide lipid molecules to model the SC and investigate the phase behavior of ceramide bilayers as a function of temperature.[7] Moreover, ternary lipid-mixture bilayer systems consisting of ceramides, cholesterols, and fatty acids have been used to more realistically model the SC in MD simulations.[14] Recently, Del Regno et al. constructed multilamellar systems composed of two ternary skin-lipid bilayers and investigated the molecular permeation pathways through the SC by MD simulations.[3]

In this study, we performed a series of all-atom MD (AAMD) simulations to assess the formation and properties of SLES micelles in aqueous solutions and to characterize and compare the behaviors of SLES micelles on ceramide and DMPC bilayer surfaces. We began by determining the MD-simulated distributions and average radii of SLES micelles in aqueous solutions and comparing them with the experimentally measured counterparts using dynamic light scattering (DLS). Events occurring when SLES micelles approached ceramide and DMPC bilayer surfaces were examined using constant-force steered MD (cf-SMD) simulations. The behaviors of SLES monomers in micelles and on ceramide and DMPC bilayer surfaces were thoroughly analyzed and identified according to variations in the Gibbs free energy plotted as the functions of the potential of mean force (PMF) constructions and decompositions.

The remaining paper is organized as follows. Computational and Experimental Methods as well as the simulation systems are described in the next section. The MD simulation results are presented, and their significance is discussed in the Results and Discussion section. Finally, the Conclusions section summarizes the main study findings and their significance in the context of the research objectives and proposes a direction for further work.

COMPUTATIONAL AND EXPERIMENTAL METHODS

Simulation Systems. The surfactant selected for this work was sodium laureth sulfate showing two OE groups (SLE2S) (Figure 1c). To assess the thermodynamic behaviors of SLE2S in water and in lipid bilayers, we prepared six systems, whose details are provided in Table 1. System 1 was prepared to assess the formation and size distribution of SLE2S micelles in aqueous solutions (Figures 1d and S1a). The concentration of

Table 1. Details of Six Simulation Systems

	N						
system	SLE2S[a]	Cer240	DMPC	water	*P* (bar)	*T* (K)	simulation time (ns)
1[b]	100			39 600	1.0	298.15	112[c]
2	16	256		19 737	1.0	350.15	400[d]
3	16		256	29 377	1.0	313.15	400[d]
4	16			13 858	1.0	350.15 313.15	30[e]
5	1	150		7780	1.0	350.15	30–80[f]
6	1		128	7780	1.0	313.15	30–80[f]

[a]Sodium cations also included as counterions to neutralize system. [b]Approximately 5.0% (w/v) or 0.133 M aqueous SLE2S solution was used in DLS experiments. [c]A 110 ns equilibration run in NPT ensembles followed by a 2 ns production run in NVE ensembles. [d]Total cf-SMD simulation time. [e]Total simulation time in window consisting of 10 and 20 ns equilibration and production runs, respectively. [f]Total simulation time in window consisting of 10–60 and 20 ns equilibration and production runs, respectively.

https://dx.doi.org/10.1021/acs.jpcb.0c02856
J. Phys. Chem. B 2020, 124, 5919−5929

SLE2S molecules in system 1 exceeded the critical micelle concentration (CMC) of the chosen SLES surfactant[34] and was approximately 5% (w/v) (≈0.133 M) of the SLE2S solution used in DLS experiments to measure SLE2S micelle sizes. The behaviors of an SLE2S micelle on bilayer surfaces were characterized and compared using cf-SMD simulations in systems 2 and 3, wherein a micelle consisting of 16 SLE2S monomers was placed 50 Å away from the center of a prepared ceramide or DMPC bilayer in the z direction, that is, normal to the bilayer surface (Figure 1e and f). The micelle used in systems 2 and 3 was selected from among 41 micelles obtained from six independent 110 ns MD trajectories for system 1 (Figure S3). In system 4, the same SLE2S micelle as the one used in systems 2 and 3 was placed in a water box to generate PMF profiles for the transfer of a SLE2S surfactant from the micelle to bulk water (Figures 1g and S1b). Systems 5 and 6 were constructed to acquire PMF profiles for the partitioning of a SLE2S surfactant from bulk water to ceramide or DMPC bilayer (Figure 1h and i). The CHARMM general force field (CGenFF) was used for the SLE2S surfactant model,[35,36] and the CHARMM36 and TIP3P water models were used for ceramide and DMPC lipid molecules and for water, respectively.[37,38] The isothermal−isobaric (NPT) ensemble was applied for all six systems, and periodic boundary conditions were utilized in x, y, and z directions. The Langevin thermostat method was employed to maintain temperatures at 298.15, 350.15, and 313.15 K for systems 1 and 4, 2 and 5, and 3 and 6 with a damping coefficient of 5 ps^{-1}. Pressure was controlled isotropically for systems 1 and 4 and anisotropically for the other systems at 1 bar, using the Nosé−Hoover Langevin piston method.[39] Particle-mesh Ewald (PME) summation was used to calculate electrostatic interactions, while van der Waals (vdW) interactions were modeled using the Lennard-Jones potential, wherein a switching function showing cutoff and switching distances of 12 and 10 Å, respectively, was applied. All of the systems were neutralized with the same number of sodium cations as that of the SLE2S surfactant anions. Ceramide bilayers for systems 2 and 5 were generated in a hexagonal lattice structure, above and below which water molecules were subsequently placed.[40] After the system energy was minimized, hydrated ceramide bilayer systems were initially equilibrated using the NVT ensemble at 300.15 K for 10 ns. After re-equilibration for 50 ns under semi-isotropic conditions of 1 bar and 300.15 K, the ceramide bilayer-system temperatures were gradually increased by 10 K per 20 ns under the same pressure condition, finally reaching 350.15 K where the system was re-equilibrated for 20 ns under anisotropic conditions of 1 bar. The average lipid area and bilayer thickness (defined as the distance between peaks in total-electron-density profile in Figure S2) were 47.01 $Å^2$ and 46.2 Å, respectively, for the hydrated ceramide bilayer system in this work, which are in good agreement with previous simulation studies.[14] Hydrated DMPC bilayer systems for systems 4 and 6 were prepared using the CHARMM-GUI Membrane Builder (www.charmm-gui.org)[41,42] and were equilibrated for 5 ns under semi-isotropic conditions at 1 bar and 313.15 K and then re-equilibrated for 100 ns under anisotropic conditions at the same temperature and pressure. All ceramide bilayer simulations were performed at 350.15 K because the ceramide bilayer system remains in an ordered-gel phase wherein the physiological skin condition is maintained and bilayer structural and thermodynamic properties only slightly vary between 300 and 350 K, while system dynamics accelerate at higher temperatures.[7,10] All of the MD simulations were performed using the NAMD program package with a time step of 2 fs, and all systems were visualized using the VMD program package.[43,44]

Determination of SLE2S-Micelle Size (System 1). Six independent 110 ns MD simulations were performed for system 1 to acquire a statistically meaningful size for SLE2S micelles in aqueous solutions at 1 bar and 298.15 K. To define a micelle, we first selected a single SLE2S surfactant and calculated the distance between each carbon atom in the tail group of the selected surfactant and carbon atoms in the tail group of another surfactant. If the distance is within 5 Å, we assumed that the two surfactants belong to the same micelle. We repeated this process for all of the surfactants in the simulation box. A total of 41 SLE2S micelles were obtained from six independent simulation trajectories, and three different methods were used to calculate the micellar distribution and average radius (Figure S3). The first method combines the Einstein−Smoluchowski and Stokes−Einstein relations. Initially, the diffusion coefficient of each micelle was evaluated using the Einstein−Smoluchowski diffusion formula

$$6D^{\mathrm{M}}t = \langle (\mathbf{R}^{\mathrm{M}}_{\mathrm{COM}}(t) - \mathbf{R}^{\mathrm{M}}_{\mathrm{COM}}(0))^2 \rangle_t \tag{1}$$

where $\langle \cdots \rangle_t$ means the time average, D^{M} represents the micelle diffusion coefficient, and $\langle (\mathbf{R}^{\mathrm{M}}(t) - \mathbf{R}^{\mathrm{M}}(0))^2 \rangle$ is the mean square displacement (MSD) of the micellar center of mass (COM).[45] Then, the hydrodynamic radius of a micelle, $R^{\mathrm{M}}_{\mathrm{h}}$, can be estimated using the diffusion coefficient obtained in eq 1 and the Stokes−Einstein relation:

$$R^{\mathrm{M}}_{\mathrm{h}} = \frac{k_{\mathrm{B}}T}{6\pi\eta_{\mathrm{w}}D^{\mathrm{M}}} \tag{2}$$

Here, k_{B} is the Boltzmann constant, T is the temperature, and η_{w} is the viscosity of water (=8.91 × 10^{-4} kg/m/s at 25 °C).[46] Diffusion coefficients were calculated for 41 SLE2S micelles by conducting additional 2 ns microcanonical ensemble (NVE) simulations after 110 ns runs for each trajectory (Figure S4).[47] In this work, diffusion coefficients were not corrected to account for finite simulation-box size because an approximately 107 Å simulation box in system 1 is much larger than an estimated 10−15 Å SLE2S micelle radius.[48,49] In the second method, the root-mean-square radius of gyration for each micelle, $R^{\mathrm{M}}_{\mathrm{g}}$, was calculated using the equation[50]

$$R^{\mathrm{M}}_{\mathrm{g}} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\langle (\mathbf{R}^{i}_{\mathrm{COM}} - \mathbf{R}^{\mathrm{M}}_{\mathrm{COM}})^2 \rangle_t} = \sqrt{\frac{1}{2N^2}\sum_{i=1}^{N}\sum_{j=1}^{N}\langle (\mathbf{R}^{i}_{\mathrm{COM}} - \mathbf{R}^{j}_{\mathrm{COM}})^2 \rangle_t} \tag{3}$$

where N is the number of SLE2S monomers in a single micelle, $\mathbf{R}^{\mathrm{M}}_{\mathrm{COM}}$ is the COM of that micelle, and $\mathbf{R}_i$ and $\mathbf{R}_j$ are the COM positions of the ith and jth monomers in that micelle, respectively (Figure S5). In the third method, the micellar radius is estimated from the average distance between the micellar sulfur-atom positions and the COM, which is written as

$$R^{\mathrm{M}}_{\mathrm{s}} = \frac{1}{N}\sum_{i=1}^{N}\langle \sqrt{(\mathbf{R}^{i}_{\mathrm{S}} - \mathbf{R}^{\mathrm{M}}_{\mathrm{COM}})^2} \rangle_t \tag{4}$$

https://dx.doi.org/10.1021/acs.jpcb.0c02856
J. Phys. Chem. B 2020, 124, 5919−5929

where $\mathbf{R}_S^i$ means the position of a sulfur atom in the ith SLE2S surfactant in a micelle (Figure 1c).

Steered MD Simulations for SLE2S Micelle Interactions with Ceramide and DMPC Lipid Bilayers (Systems 2 and 3). To describe the interactions of SLE2S micelles with ceramide and DMPC bilayers, we performed cf-SMD simulations for systems 2 and 3. Prior to simulations, a single 16-SLE2S-monomer micelle was placed 50 Å above the ceramide or DMPC bilayer center. When the micelle in each system was restrained using a half-harmonic constraint with a force constant of 1.0 kcal/mol/Å^2, both systems were equilibrated for 5 ns under semi-isotropic NPT conditions at 1 bar and 350.15 K (system 2) or 313.15 K (system 3) and then re-equilibrated for 5 ns under anisotropic NPT conditions at the same pressure and temperature. Then, cf-SMD simulations were performed for 400 ns for both systems to pull SLE2S micelles down to the bilayer surfaces under the given pressure and temperature conditions, using a constant force with a weak force constant of 0.5 kcal/mol/Å^2 applied to the micelles in the negative z direction.

Generation of PMF Profiles for SLE2S Monomer Transferring from Micelle to Water and Subsequently to Ceramide and DMPC Bilayers (Systems 4, 5, and 6). Changes in Gibbs free energy for an SLE2S surfactant transferring from a micelle to ceramide and DMPC bilayers were indirectly determined by calculating PMF profiles for an SLE2S monomer transferring from a 16-mer micelle to bulk water (system 4) and then to ceramide and DMPC bilayers (systems 5 and 6). PMF profiles were generated using umbrella sampling (US) postprocessed by the weighted-histogram analysis method (WHAM).[51−53] The reaction coordinate for system 4, defined as distance d between the COM of a single micelle SLE2S monomer and micellar COM, ranges from 5.0 to 30.0 Å and was divided by 51 windows with a width of 0.5 Å. For systems 5 and 6, the reaction coordinates were set as distances ranging from 0.0 to 40.0 Å in the z direction between an SLE2S-monomer COM and the center of the ceramide or DMPC bilayer. The window width was 0.5 Å, leading to a total of 81 windows. The initial configuration for each window in the three systems was obtained by placing an SLE2S monomer at the center of the window. Then, a harmonic window potential showing a force constant of 4.0 kcal/mol/Å^2 was applied to restrain the SLE2S monomer in that window. In system 4, COMs of the other 15 monomers except the one placed in the US window were maintained within 17 Å inside the micelle by a flat-bottom half-harmonic restraint with a force constant of 0.5 kcal/mol/Å^2. SLE2S monomer dynamics were slower when the monomer COM was placed at the interface between the micelle and water region or between the bilayer and water region than when it remained completely in one domain, such as only in the micelle, bilayer, or water region. As a result, depending on the SLE2S monomer position, the simulation time for a single US trajectory in each window varied from 30 to 80 ns, during which the first 10−60 ns and last 20 ns were used as equilibration and production runs, respectively. Two PMF profiles were generated for system 4; one at 1 bar and 350.15 K and the other at 1 bar and 313.15 K. The PMF profile generated for system 5 was calculated at 1 bar and 350.15 K, while the one generated for system 6 was obtained at 1 bar and 313.15 K. Thermodynamic integration (TI) was also employed to decompose the PMF profiles into individual components to determine the influence of each system component on the Gibbs free energy of SLE2S monomer transfer.[53−55]

Determination of SLE2S Micelle Size by DLS. SLE2S-micelle size distributions were experimentally determined by DLS measurements using a Malvern Zetasizer Nano ZSP instrument (www.malvernpanalytical.com). SLE2S surfactant concentrations were 5.0% (w/v) aqueous, and the micelle sizes were measured at 25 °C and 1.0 atm (Figure S6).

RESULTS AND DISCUSSION

Distribution profiles for SLE2S micelle radii evaluated by the three independent methods mentioned in the previous section are illustrated in Figure 2. In addition, average SLE2S-micelle

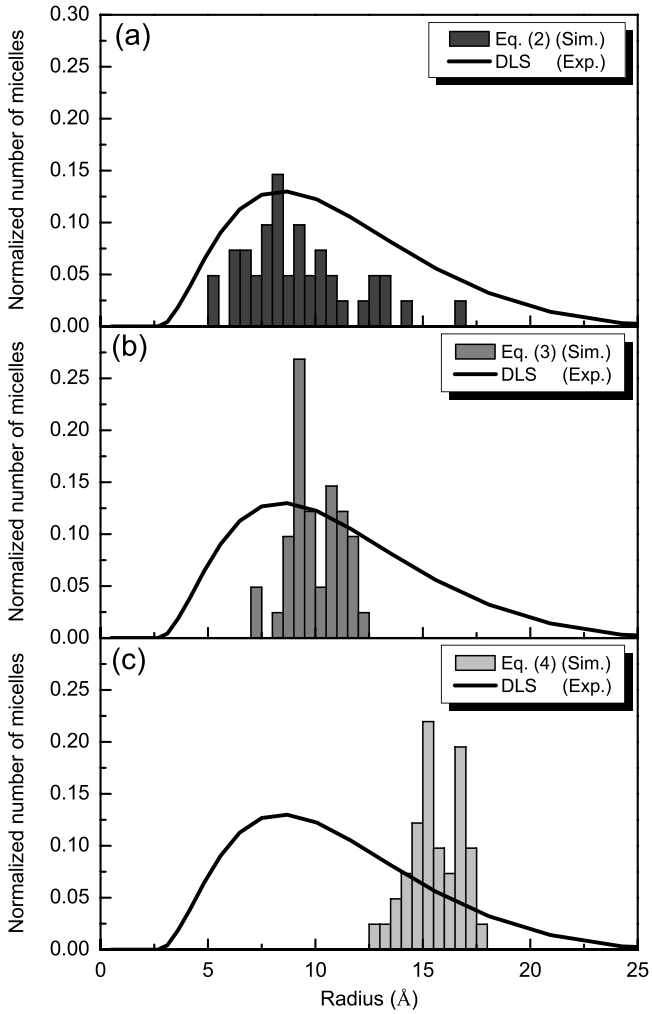


Figure 2. Distribution profiles for SLE2S micelle radii evaluated by (a) diffusion coefficient calculations, (b) gyration radii, and (c) distances between sulfur-atom positions and micelle COMs in six MD simulations for system 1 and comparisons with distributions obtained from DLS experiments. Distribution profiles are normalized such that the summations or integrations over the number of micelles become unity.

aggregation numbers, diffusion coefficients, and radius distributions obtained from MD simulations for system 1 and DLS experiments are presented in Table 2. Table 2 and Figure S7 show that the aggregation number, that is, the number of SLE2S monomers in a micelle formed during 110 ns, ranges widely from 6 to 24 with an average of $\langle N \rangle$ = 14.6.

https://dx.doi.org/10.1021/acs.jpcb.0c02856
J. Phys. Chem. B 2020, 124, 5919−5929

Table 2. Average Values for 41 SLE2S Micelles Obtained from Six Independent Simulations for System 1[a]

			micelle radius (Å)			
	$\langle N_{SLE2S}{}^{b}\rangle^{c}$	$\langle D^{M}\rangle$ ($\times 10^{-2}$ Å^2/ps)	$\langle R_h^M\rangle$	$\langle R_g^M\rangle$	$\langle R_s^M\rangle$	$\langle R_{DLS}^M\rangle$
avg. value	14.6	2.85	9.20	10.0	15.6	9.23
std. error	±0.70	±0.12	±0.40	±0.24	±0.12	

[a]Information on each of the 41 micelles is provided in Table S1. [b]N_{SLE2S} is the aggregation number, that is, the number of SLE2S monomers in a micelle. [c]$\langle\cdots\rangle$ is the average value over the 41 micelles.

Average aggregation numbers of SLE2S surfactants are substantially lower than those of SDS micelles because the CMC of SDS (c_{SDS}^{CMC} = 8.2 mM) is approximately 10 times higher than that of SLE2S (c_{SLE2S}^{CMC} = 0.80 mM).[34,56] A previous study suggested that SLE2S surfactants (which have two oxyethylene (OE) groups added to SDS) show more reduced electrostatic repulsive interactions between anionic-sulfate heads owing to attractive intramolecular or intermolecular ion−dipole interactions between sulfate heads and adjacent OE groups, leading to increased hydrophobicity.[34] In the MD simulations for system 1, the distribution and mean radius of SLE2S micelles evaluated on the basis of micelle diffusion coefficients show better agreement with DLS measurements than with the other method based on the gyration radius or distance between the sulfur atoms and micellar COM mainly because the DLS-estimated micelle radius is also based on the Einstein−Smoluchowski and Stokes−Einstein relations. The ratio between the radius of gyration and hydrodynamic radius, $\rho = R_g/R_h$, is one of the key factors to determine the shape of micelles or polymer.[57] The evaluated ratio $\rho^M = \langle R_g^M\rangle/\langle R_h^M\rangle$ = 1.09 for the SLE2S micelle in Table 2 is between ρ_{sp} = 0.78 for the spherical shape and ρ_{rc} = 1.5 for the random-coil, indicating that the shape of SLE2S micelles seems to deviate from a spherical geometry (Figure S8a).[58] This can be in part attributed to a relatively broad distribution of the aggregation number in the micelle shown in Figure S7 because the wide distribution indicates an increase in the polydispersity of the micelles and the number of nonspherical micelles (Figure S8b).[59] Another reason for the deviation can be due to an increase in the micelle diffusion coefficient as a result of electrostatic couplings between negatively charged micelles and counterions to maintain electroneutrality in high concentrations.[60,61] The average radius of gyration ($\langle R_g^M\rangle$ = 10.0 Å) is smaller than that estimated from the distance between sulfur atoms and the micelle COM ($\langle R_s^M\rangle$ = 15.6 Å) because the micelle radius of gyration is defined in eq 3 as the average distance between the COMs of all of the surfactants in a micelle and the micellar COM itself and because SLE2S surfactant COMs in the micelle are closer to the micellar COM than the head sulfur atoms are (see Figure 1c).

Representative snapshots from 400 ns cf-SMD simulations for a 16mer SLE2S micelle approaching ceramide and DMPC lipid bilayer surfaces are illustrated in Figures 3 and 4. A weak pulling force showing a force constant of 0.5 kcal/mol/Å^2 was applied to the micelle to pull it down to the lipid bilayer surface without considerable disturbance. Lipid-tail conformations and orientations shown in Figures 3 and 4 indicate that the ceramide bilayer is in an ordered-gel phase at 1 bar and 350.15 K, while the DMPC bilayer remains in a disordered-fluid phase at 1 bar and 313.15 K. In Figures 3a and S9, a small opening is generated on the upper-leaflet surface where several micelle surfactants are reoriented, and the micelle is slightly deformed from spherical to hemispherical. SLE2S surfactant is then transferred from the micelle to the ceramide bilayer upper leaflet at ∼100 ns (Figure 3b), two more SLE2S monomers are incorporated into the bilayer at 235 ns (Figure 3c), and the three surfactants remain in the bilayer at 400 ns (Figure 3d), indicating that micellar SLE2S surfactants can be partitioned into the skin through the SC layer. Our simulation results are also supported by experimental observations that micellar surfactants show enhanced partitioning into model skin-lipid membranes compared to disaggregated surfactants.[8] It is also worth noting in Figure 3b−d that the tail orientations of surfactant monomers incorporated into the ceramide bilayer are parallel to ceramide lipid-molecule tails. Unlike ceramide bilayers, wherein several SLE2S monomers are transferred to the bilayer, SLE2S surfactants are not transferred to the DMPC bilayer up to 400 ns, as shown in Figure 4. Absence of SLE2S transfer in the DMPC bilayer can be attributed to electrostatic interactions between negatively charged SLE2S-surfactant heads and zwitterionic DMPC molecule heads, which prevent micelle surfactant monomers from reorienting from upside-down orientation to upright orientation to be inserted into the bilayer on the bilayer surface. Such electrostatic interactions are not shown in Figure 3 because ceramide molecule heads

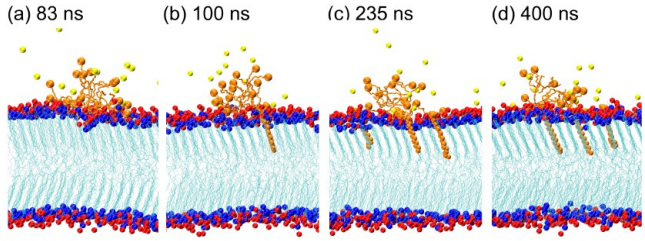


Figure 3. Representative snapshots from the 400 ns cf-SMD simulation trajectory for the 16mer SLE2S micelle interacting with the ceramide lipid bilayer at (a) 83, (b) 100, (c) 235, and (d) 400 ns, respectively. The cf-SMD simulation was conducted at 1 bar and 350.15 K. Graphical representations and atomic colors are the same as those in Figure 1. SLE2S-monomer tails partitioned into ceramide bilayer are represented by vdW spheres; the others are represented by lines.

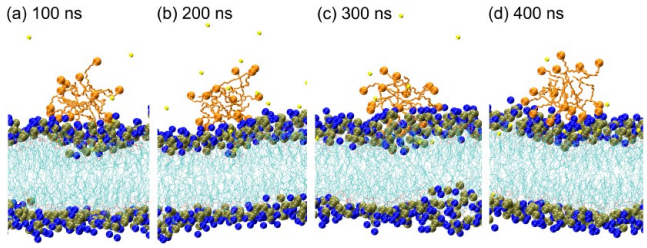


Figure 4. Representative snapshots from the 400 ns cf-SMD simulation trajectory for the 16mer SLE2S micelle interacting with the DMPC lipid bilayer at (a) 100, (b) 200, (c) 300, and (d) 400 ns, respectively. The cf-SMD simulation was conducted at 1 bar and 313.15 K. Graphical representations and atomic colors are the same as those in Figure 1.

https://dx.doi.org/10.1021/acs.jpcb.0c02856
J. Phys. Chem. B 2020, 124, 5919−5929

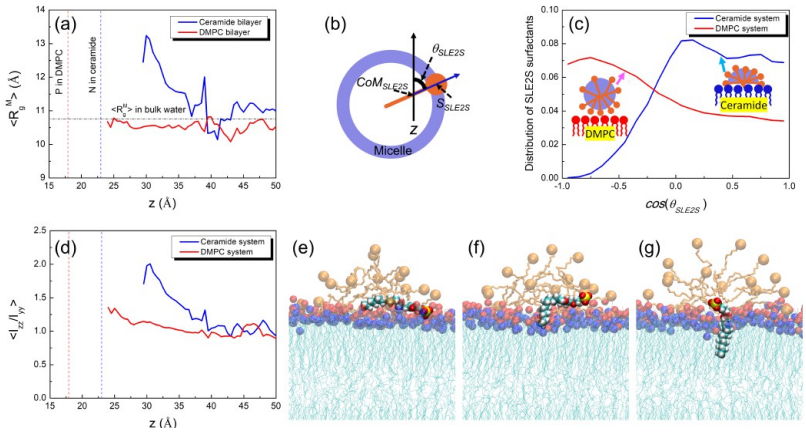


Figure 5. In systems 2 and 3, (a) changes in the radius of gyration ($\langle R_g^M \rangle$) of the micelles, (b) angle definition in the calculation of angle distributions, (c) angle distributions inside the micelles as they stay close to the bilayer surfaces, and (d) changes in the ratio between the moments of inertia (between I_{zz} and I_{xx}) of the micelles. (e−g) Snapshots representing the transfer process of a surfactant from the deformed micelle to the ceramide bilayer in system 2. In parts a and d, the properties were calculated as a function of the COM of the micelles. The insets in part c illustrate a representative configuration of SLE2S surfactants in the micelles on the surface of the ceramide and DMPC bilayers.

lack the zwitterionic character. SLE2S micelle deformation observed on the ceramide bilayer surface in Figure 3a is not shown in Figure 4, where the micellar shape appears to be maintained at the DMPC bilayer surface. Such distinct bilayer-dependent SLE2S-micelle behaviors will be addressed later with respect to changes in Gibbs free energy. To confirm that the deformation of the SLE2S micelle on the surface of the ceramide bilayer in Figure 3is not an artifact due to a weak pulling force, we prepared two systems where three SLE2S micelles were placed on either ceramide bilayers or DPMC bilayers and then performed 200 ns MD simulations for the two systems in the absence of pulling force. In the MD simulation with the ceramide bilayer, micelle deformations and surfactant transfers into the bilayer were still observed, while no such events were detected on the surface of the DMPC bilayer (Figure S10).

Micelle conformations on the bilayer surfaces in systems 2 and 3 were more systematically analyzed by calculating changes in the radius of gyration of the micelles, distributions of SLE2S surfactants inside the micelles when they stay close to the bilayer surface, and changes in the ratio of the moments of inertia of the micelles (Figure 5a−d). In addition, the process of surfactant transfer from the deformed micelle to the ceramide bilayer in system 2 is visualized in Figure 5e−g. Figure 5a illustrates that, as the SLE2S micelle approaches the ceramide bilayer surface in system 2, $\langle R_g^M \rangle$ of the micelle increases due to a conformational change in the micelle, while that quantity remains close to the value in bulk water in the case of the DMPC bilayer, being indicative of the maintenance of the spherical shape in system 3. When the micelles stay close to the bilayer surfaces in systems 2 and 3, angle distributions of surfactants inside the micelles, in which an angle is defined as the angle between a vector pointing to the sulfur atom from the COM of a surfactant and the z axis (Figure 5b), are presented in Figure 5c. The angle distributions reveal that the head groups of most surfactants in a micelle are oriented upward, indicating that the spherical shape of the SLE2S micelle in bulk water is deformed into a hemispherical shape on the ceramide bilayer surface. On the DMPC bilayer surface, however, surfactants with head groups downward are slightly more populated than the ones with head groups upward in a micelle and spherical shapes are still maintained. Insets in Figure 5c illustrate the distribution of surfactants in a micelle on the bilayer surfaces. Ratios of I_{zz} to I_{yy} are exhibited in Figure 5d where I_{aa} is defined as the moment of inertia of a micelle around the a axis (a = x, y, or z). As a micelle approaches the bilayer surface, the ratio for the micelle in the ceramide bilayer becomes higher than that for the hemisphere whose value is about 1.54, while the ratio in the DMPC bilayer slightly increases from the value of a sphere which is unity, also confirming a conformational change of the micelle on the ceramide bilayer surface. Three snapshots in Figure 5e−g also clearly demonstrate how a surfactant permeates from a deformed micelle to the ceramide bilayer on the bilayer surface.

Figure 6a shows entropy-corrected PMF profiles for an SLE2S monomer transferring from a micelle to bulk water at 350.15 and 313.15 K, plotted as functions of distance (d) between the COMs of the transferring surfactant monomer and the micelle consisting of the other 15 monomers except the transferring monomer in system 4. PMF profiles in Figure 6a were generated by correcting original PMF profiles in Figure S11 using the equation $\Delta G(d)_{\text{corrected}} = \Delta G(d)_{\text{original}} + k_B T \ln d^2$, where the second term represents entropy correction, which appears because entropy depends on the phase volume $4\pi d^2 \delta d$.[48] PMF curve minima in Figure 6a are located at d_{min} = 11.0 and 10.5 Å at 350.15 and 313.15 K, respectively, which agree well with $\langle R_g^M \rangle$ = 10 Å because $\langle R_g^M \rangle$ corresponds to the average distance between micelle−monomer COMs and the micellar COM itself. Changes in Gibbs free energy of an SLE2S surfactant transferring from a micelle to bulk water, $\Delta G_{\text{SLE2S}}^{\text{m}\to\text{w}}$, can be estimated using the CMC of SLE2S and the following equation[48,62]

$$\Delta G_{\text{SLE2S}}^{\text{m}\to\text{w}} = -k_B T \ln x_{\text{SLE2S}}^{\text{CMC}} = -k_B T \ln\left(\frac{n_{\text{SLE2S}}^{\text{CMC}}}{n_{\text{SLE2S}}^{\text{CMC}} + n_{\text{wat}}}\right) \approx -k_B T \ln\left(\frac{n_{\text{SLE2S}}^{\text{CMC}}}{n_{\text{wat}}}\right) \quad (5)$$

where $x_{\text{SLE2S}}^{\text{CMC}}$, $n_{\text{SLE2S}}^{\text{CNC}}$, $n_{\text{SLE2S}}^{\text{CMC}}$, and n_{wat} are the mole fraction and mole number of SLE2S surfactants and mole numbers of water molecules at the CMC of SLE2S, respectively. $\Delta G_{\text{SLE2S}}^{\text{m}\to\text{w}}$ was estimated as 6.93 kcal/mol at 313.15 K based on eq 5 and the

https://dx.doi.org/10.1021/acs.jpcb.0c02856
J. Phys. Chem. B 2020, 124, 5919−5929

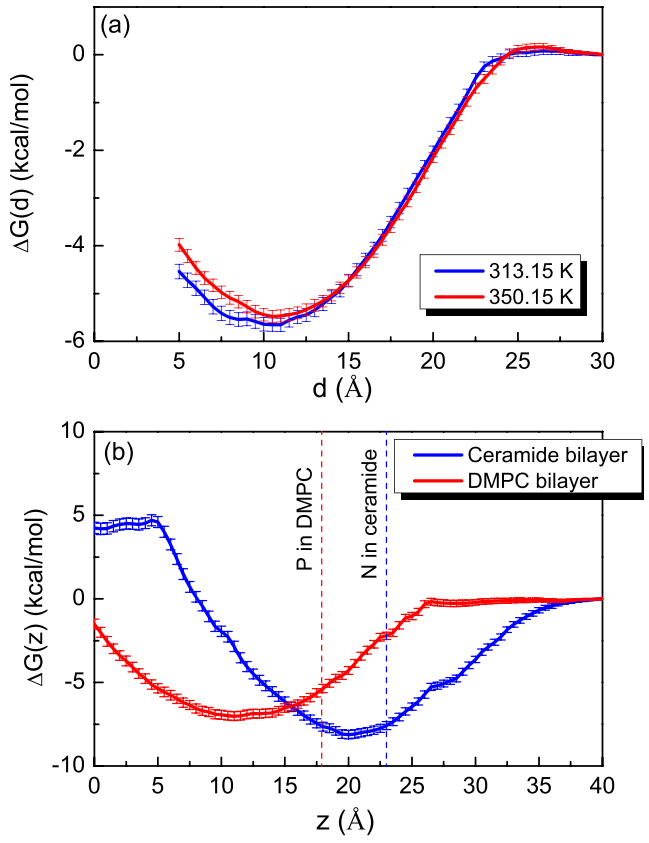


Figure 6. (a) Entropy-corrected PMF profiles plotted as the functions of the COM for the SLE2S monomer transferring from a micelle to bulk water at 350.15 and 313.15 K and (b) PMF profiles for the SLE2S monomer inserting into the ceramide and DMPC bilayers along with corresponding error bars. The micelle in part a is composed of 16 SLE2S surfactant monomers including that escaping from the micelle. In part b, ceramide and DMPC bilayer PMF profiles were obtained at 350.15 and 313.15 K, respectively, and vertical blue and red broken lines indicate average z coordinates of nitrogen (N) and phosphorus (P) atoms in ceramide and DMPC lipid molecules in bilayer upper leaflets, respectively, as shown in Figure 1a and b.

experimentally obtained SLE2S CMC of $c_{\mathrm{SLE2S}}^{\mathrm{CMC}}$ = 0.80 mM at 298.15 K, which is in fair agreement with the transfer Gibbs free energy of 5.65 kcal/mol evaluated from the minimum $\Delta G(d)$ at 313.15 K in Figure 6a.[34] Figure 6a also shows that $\Delta G_{\mathrm{SLE2S}}^{\mathrm{m}\rightarrow\mathrm{w}}$ at 350.15 K is slightly lower than that at 313.15 K, which is indirectly supported by an experimental result showing that the transfer Gibbs free energy of an SDS surfactant decreases with increasing temperature.[63]

Changes in Gibbs free energy for an SLE2S surfactant monomer transferring from bulk water to a ceramide or DMPC bilayer are illustrated in Figure 6b. The decrease in Gibbs free energy outside the bilayer begins farther from the bilayer surface and is larger for the ceramide bilayer than for the DMPC counterpart, and the minimum Gibbs free energy inside the bilayer is slightly lower for ceramide than for DMPC, indicating that the SLE2S surfactant is more strongly attracted to the ceramide bilayer than to the DMPC bilayer. A slight decrease in the PMF profile for a surfactant at the center of the ceramide bilayer can be attributed to the fact that the lipids are not so tightly packed at the bilayer center due to the mismatch in the length of the two hydrocarbon chains of the ceramide (Figure S2).[5] SLE2S micelle deformation observed on the ceramide bilayer surface in Figure 5 can reduce the Gibbs free energy barrier for a surfactant exiting out of a micelle. In addition, the attractive Gibbs free energy curve for surfactant insertion into the ceramide bilayer (Figure 6b) demonstrates that SLE2S surfactants can partition from a micelle into the ceramide bilayer, as displayed in Figure 3. In contrast with micelle deformation on the ceramide bilayer surface, such deformation does not occur on the DMPC bilayer surface, leading to higher Gibbs free energy barrier for an SLE2S surfactant transferring from a micelle to bulk water. Compared to variations in Gibbs free energy for surfactant insertion into the ceramide bilayer, the less attractive Gibbs free energy curve for surfactant insertion into the DMPC bilayer and the higher Gibbs free energy barrier for transferring the surfactant from the micelle to bulk water imply that the transfer of an SLE2S surfactant from a micelle to the DMPC bilayer is highly unlikely, as already shown in Figure 4.

PMF profiles shown in Figure 6b were decomposed into system components presented in Figures 7 and S12, revealing

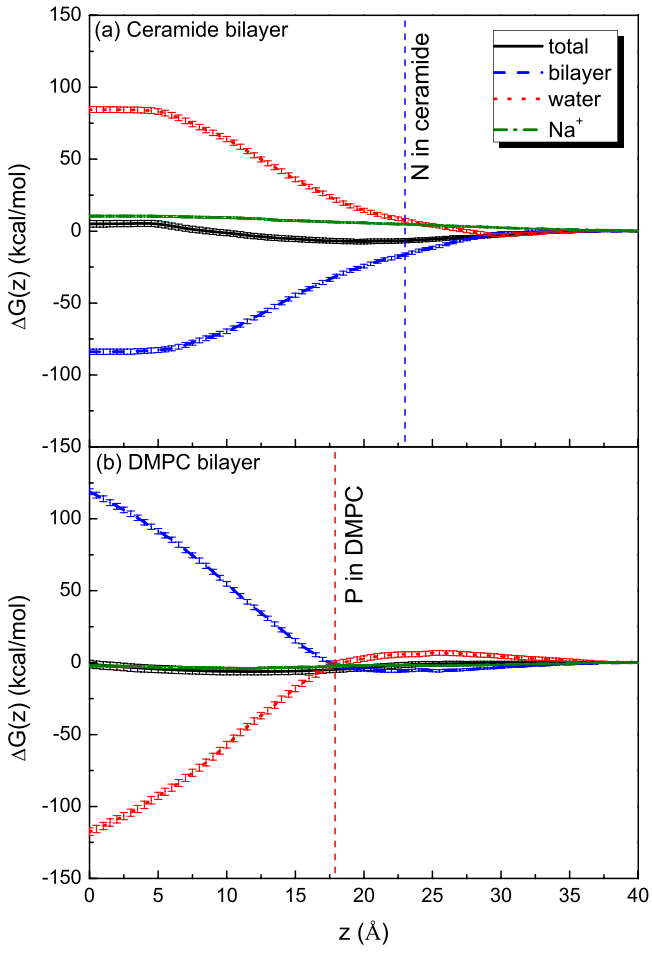


Figure 7. PMF profiles for curves in Figure 6b decomposing into components in (a) ceramide and (b) DMPC bilayer systems. Vertical blue and red broken lines indicate the same as that in Figure 6b.

that the decomposition profiles are bilayer-dependent and that overall PMF decomposition profiles are determined almost exclusively by the interactions of surfactant hydrophilic heads with system components (systems 5 and 6). Figure 7a shows that, as an SLE2S surfactant externally approaches the ceramide bilayer surface, the Gibbs free energy of the monomer interaction with the ceramide bilayer and water

https://dx.doi.org/10.1021/acs.jpcb.0c02856
J. Phys. Chem. B 2020, 124, 5919−5929

slightly decreases, which may be attributed to favorable interactions of the SLE2S-monomer hydrophilic head with ceramide hydrophilic heads and water molecules between them. Inside the ceramide bilayer, however, the SLE2S monomer interacts with the ceramide bilayer and water molecules in a completely different way. As shown in Figure 7a and in Figure S12a and b, the SLE2S monomer inside the bilayer has favorable interactions with the ceramide bilayer probably owing to hydrophilic interactions between surfactant and ceramide heads, while the interaction of surfactant with water molecules becomes unfavorable owing to the dehydration penalty associated with the monomer anionic head inside the bilayer. In contrast with the SLE2S surfactant in the ceramide bilayer, Figure 7b and Figure S12c and d show that the surfactant inside the DMPC bilayer has highly favorable interactions with water molecules because the SLE2S surfactant keeps its anionic head externally oriented to maintain contact with water molecules in the boundary between the head and tails of the DMPC bilayer. However, interactions between the surfactant and the DMPC bilayer become unfavorable owing to water molecule interference in the DMPC bilayer head−tail boundary.

Finally, the numbers of hydrogen bonds formed between the SLE2S-surfactant head and other ceramide (system 5) and DMPC bilayer (system 6) components were calculated while the surfactant was partitioning into the bilayers and are illustrated in Figure 8, where hydrogen-bond formation is defined as in ref 55 and oxygen atoms in surfactant head sulfate and OE groups act as hydrogen-bond acceptors.[64,65] Figure 8a clearly shows that, although the number of SLE2S-surfactant and water-molecule hydrogen bonding interactions decreased, surfactant inside the ceramide bilayer can be stabilized through hydrogen-bonding interactions between surfactant heads and ceramide lipid molecules, where ceramide head hydroxyl and N−H groups act as hydrogen-bond donors. Such hydrogen-bonding interactions between surfactant and lipid molecules cannot be established in the DMPC bilayer because no hydrogen-bond donors exist in the DMPC molecule (Figure 8b). Instead, the hydrogen-bond network is still maintained between surfactant and water molecules even when the surfactant is inserted into the DMPC bilayer owing to water molecules in the DMPC bilayer head−tail boundary.

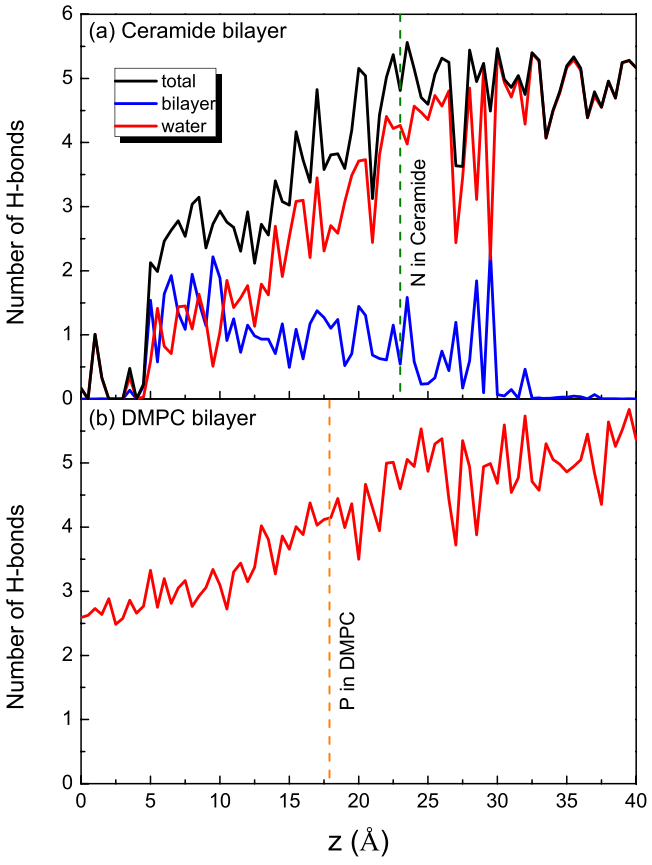


Figure 8. Number of hydrogen bonds formed between the sulfate group in the SLE2S-surfactant head and lipid and water molecules in (a) ceramide bilayer (system 5) and (b) DMPC bilayer (system 6). Hydrogen bond formation is defined in ref 52.

CONCLUSIONS

We used MD simulations and the US method to demonstrate transfer behavior and variations in Gibbs free energy for an SLE2S surfactant partitioning from a micelle into a ceramide bilayer. The radius distribution profile obtained using MSD calculations and the Stokes−Einstein relation for SLE2S micelles in water was in excellent agreement with that obtained from DLS measurements probably because computational and experimental results both are based on identical theoretical methods. On the contrary, distributions evaluated based on gyration radius or distance between micelle sulfur atoms and the micellar COM overestimate DLS measurements. When an SLE2S micelle was pulled down to the ceramide or DMPC bilayer surface in cf-SMD simulations, several SLE2S surfactants were transferred from the micelle to the ceramide bilayer, while none were transferred from the micelle to the DMPC bilayer. PMF profiles generated for the SLE2S surfactant transferring from a micelle to bulk water to a ceramide or DMPC bilayer revealed attractive interactions between an SLE2S monomer and ceramide lipid molecules through head-based electrostatic and hydrogen-bonding interactions and hydrocarbon-tail-based hydrophobic interactions, providing sufficient energy for an SLE2S surfactant to transfer from a micelle into the ceramide bilayer. In addition, SLE2S-micelle deformation on the ceramide bilayer surface helped an SLE2S surfactant to escape from the micelle by reducing the energy barrier required for exiting. However, the DMPC bilayer surface did not show any SLE2S-micelle deformation, and the energy well required for SLE2S to transfer from bulk water to the DMPC bilayer was not deep enough; therefore, the micelle surfactant was unable to obtain the energy required to overcome the energy barrier for exiting the micelle and partitioning into the bilayer. PMF decomposition profiles and hydrogen-bonding calculations revealed that an SLE2S monomer is stabilized inside the ceramide bilayer owing to hydrogen bonds and favorable interactions between the hydrophilic SLE2S surfactant head and ceramide molecules. Although we indirectly demonstrated variations in Gibbs free energy for an SLE2S surfactant partitioning from a micelle to the ceramide bilayer, we believe that our study findings will offer insight into how micellar SLE2S surfactants interact with and irritate the skin and will help in developing a surfactant that can be utilized to enhance skin penetration in transdermal delivery systems or as a solubilizer in manufacturing cosmetics. We plan to extend our computational work to design a surfactant that will alleviate or at least not exacerbate atopic dermatitis.

https://dx.doi.org/10.1021/acs.jpcb.0c02856
J. Phys. Chem. B 2020, 124, 5919−5929

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.jpcb.0c02856.

Initial configurations for systems 1 and 4 (Figure S1); total-electron-density profile for the ceramide bilayer (Figure S2); representative snapshots from six MD simulations for system 1 (Figure S3); SLE2S-micelle mean-squared displacement (MSD) profiles (Figure S4); time-evolution profiles of SLE2S-micelle gyration radii (Figure S5); DLS-measured SLE2S-micelle radius distributions (Figure S6); distribution profile for SLE2S-micelle aggregation numbers (Figure S7); ratio of radius of gyration to hydrodynamic radius for micelles (Figure S8); representative snapshot from 400 ns cf-SMD simulation for system 2 (Figure S9); snapshots for 200 ns MD simulations for three SLE2S surfactants in ceramide and DMPC bilayers (Figure S10); original PMF profiles for an SLE2S transfer from a micelle to water (Figure S11); PMF decomposition profiles for PMF curves for systems 5 and 6 (Figure S12); representative snapshots for system 5 (Figure S13); and representative snapshots for system 6 (Figure S14) (PDF)

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Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

This research was supported by the Basic Science Research Program through the National Research Foundation (NRF) of Korea (Grant No. NRF-2017R1D1A3B03028669 and 2020-R1I1A3A04037513) funded by the Korean Government (MEST). This study was conducted with support of a research grant from Kangwon National University in 2019.

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